

VIRTUAL INTERNSHIP PROJECT REPORT

Data Engineering with Python & AI/LLMs

Building Modern ELT Pipelines, Automated Ingestion Frameworks & Self-Healing Schemas
using dlt & AI Tools

Presenter
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Degree & Department
B.Tech CSE (2024-2028)

Institution IILM University,
Greater Noida

Candidate Profile & Internship Details

Student Information

Name: Pratyush Khatri

Roll Number: 2410031011

Program: B.Tech in Computer Science and Engineering

Academic Batch: 2024 – 2028

Department: School of Computer Science & Engineering

University: IILM University, Greater Noida, U.P.

Internship Overview

Course Domain: Data Engineering with Python and AI

Program Organizers: freeCodeCamp & dltHub

Course Instructors: Alexey Grigorev & Adrian Brudaru

Credential Awarded: Master of ELT Best Practices

Certificate ID: 6a940a9209b7ae6ed40a49fa

Assessment Date: August 30, 2026

Organization & Platform Profiles



dltHub (dlt - Data Load Tool)

dltHub is the primary maintainer of **dlt**, an open-source Python library designed to simplify modern data loading into production stores.

It automates complex data engineering tasks including automatic schema inference, nested JSON unwrapping, state management, and incremental ELT pipelines.

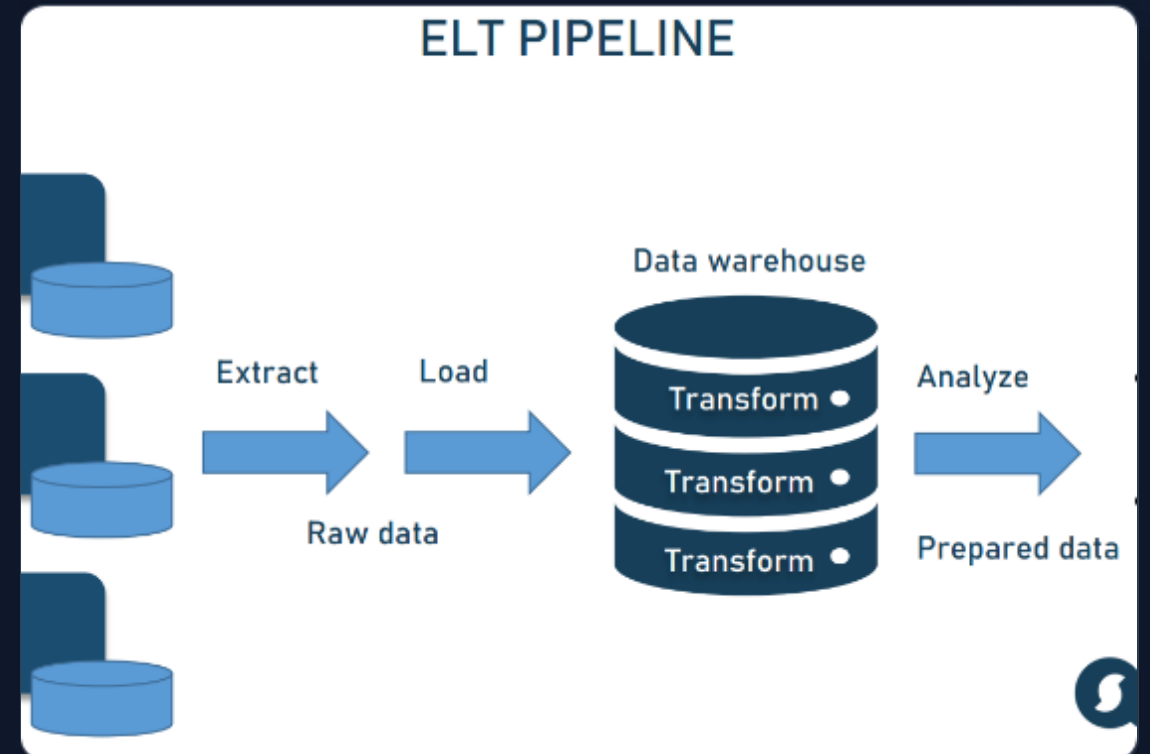
</> freeCodeCamp

freeCodeCamp is a globally recognized non-profit organization offering open-access software engineering education and industry certifications.

Through hands-on tutorials and practical project courses, it empowers millions of developers worldwide to master modern data engineering stacks.

Problem Statement: Legacy ETL Bottlenecks

- ⚠️ **API Schema Drift:** External REST APIs change payloads frequently, causing traditional downstream database pipelines to break.
- 🔗 **Nested JSON Complexity:** Manually unwrapping nested arrays requires hundreds of lines of brittle boilerplate code.
- 🚂 **Memory Management Overhead:** Fetching large batches without generator-based streaming triggers Out-Of-Memory (OOM) failures.
- 🔄 **Incremental State Tracking:** Implementing custom high-watermark cursors and backfills is error-prone and labor-intensive.



Core Project Objectives



API Extraction

Extract data from REST APIs managing authentication, pagination, rate-limiting, and memory-safe generator streams.



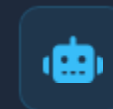
Schema Governance

Enforce Data Contracts (evolve, freeze, discard) and trigger automatic Slack alerts upon schema changes.



SCD2 & Incremental

Configure stateful incremental loads, backfilling strategies, and Slowly Changing Dimensions (SCD Type 2).



AI Vibe-Coding

Utilize OpenAI LLMs and Cursor IDE to generate pipeline prompts, write docs, and automate ETL code creation.

Technology Stack & Tools



Core Stack

Languages: Python

Libraries: Requests, Pandas, PyArrow, ConnectorX



ETL & Ingestion

Framework: dlt (Data Load Tool)

OLAP Engine: DuckDB

SQL Source: MySQL



Lakehouses & Warehouses

Cloud Warehouses: BigQuery, Snowflake, Redshift

Data Lakes: AWS S3, Parquet, Apache Iceberg, Delta Lake



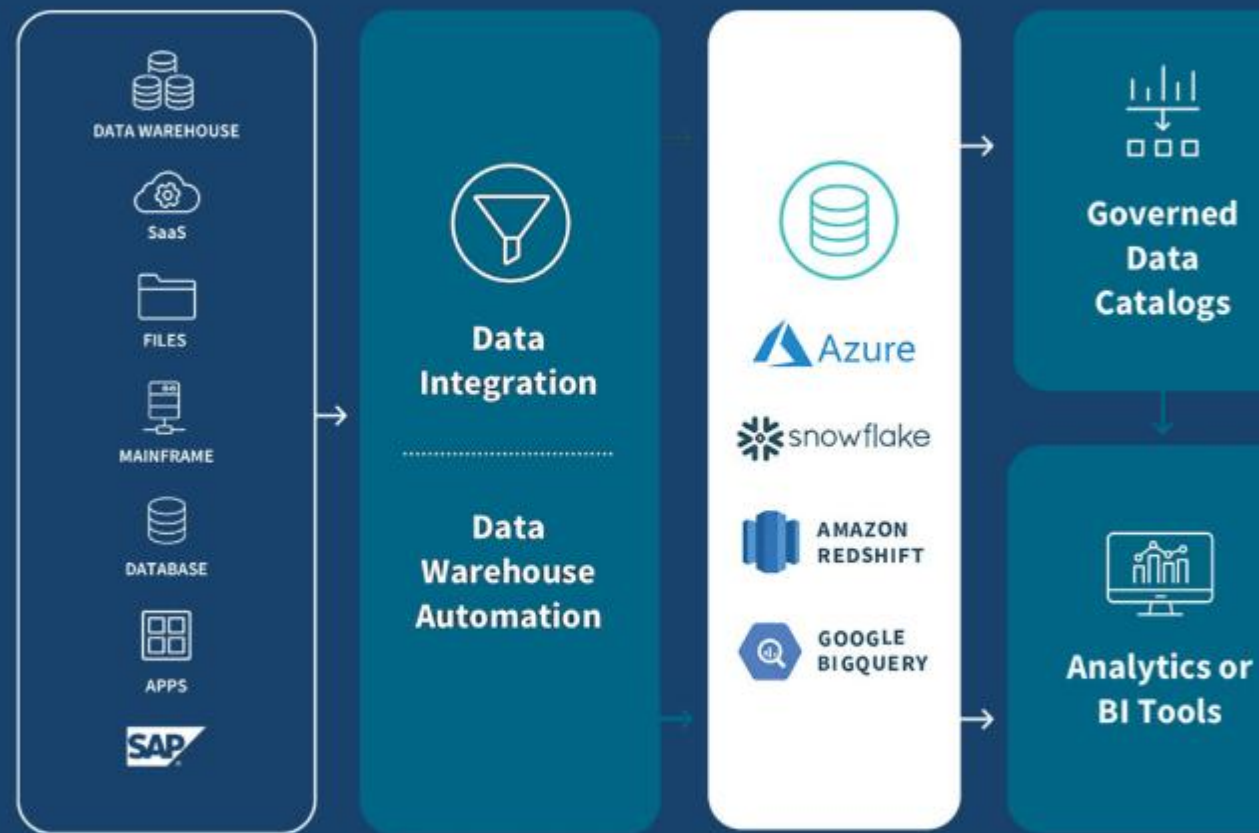
Orchestration & AI

Orchestration: Airflow, Dagster, GitHub Actions, Cron

AI Tools: OpenAI GPT, Cursor IDE

5-Layer System Architecture

-  **Source Layer:** REST APIs, relational SQL databases, raw file feeds.
-  **dlt Ingestion Layer:** Memory streaming, credentials (`.dlt/secrets.toml`), schema evolution.
-  **Normalization Layer:** Auto nested JSON flattening, SQL-to-SQL copying, SCD2 tracking.
-  **Destination Layer:** DuckDB, BigQuery, Snowflake, AWS S3 (Iceberg/Parquet).
-  **Orchestration & AI Layer:** Airflow, Dagster, GitHub Actions, LLM-generated pipelines.



Curriculum Methodology & Course Breakdown

Module Topic	Key Technical Focus & Highlights
1-3. API & Ingestion	Core data engineering principles, batch vs. streaming, REST API pagination, rate-limiting & memory safety.
4-6. Normalization & dlt	Automatic JSON flattening, dynamic schema management, DuckDB integration, configuring dlt resources & secrets.
7-8. Contracts & Incremental	Data contract enforcement (freeze/evolve/discard), Slack alerts, write dispositions, backfilling & SCD Type 2.
9-10. Tuning & Lakehouses	Parallel worker tuning, async requests, Parquet storage, AWS S3, Apache Iceberg, Delta Lake & BigQuery.
11-12. Deployment & AI	Pipeline deployment via GitHub Actions, Cron, Dagster & Airflow; LLM prompt engineering & vibe-coding in Cursor.

Expected Outcomes & Capabilities Gained

- ✔ **Self-Healing Pipelines:** Ability to build scalable, automated Python ELT data pipelines using ``dlt``.
- ✔ **Zero-Boilerplate Normalization:** Complete elimination of manual JSON flattening code through dynamic relational structuring.
- ✔ **Data Quality Governance:** Enforced schema stability through strict Data Contracts and automated alert hooks.
- ✔ **Historical Auditability:** Practical implementation of stateful incremental loads, backfills, and SCD Type 2 dimensions.
- ✔ **Multi-Cloud Deployment:** Seamless deployment across cloud lakehouses (AWS S3/Iceberg), BigQuery, and Airflow orchestrators.
- ✔ **AI Productivity Boost:** Accelerated pipeline design and documentation using OpenAI GPT models and Cursor IDE.

Credential Verification & Certification



Certified Master of ELT Best Practices

Certificate Details

Course Title: Freecodecamp: Data Engineering with Python and AI

Issued To: Pratyush Khatri (Roll No: 2410031011)

Certificate ID: [6a940a9209b7ae6ed40a49fa](#)

Date of Issue: August 30, 2026

Issuing Authority: Adrian Brudaru (Co-founder & CDO, dltHub)

Validation Status: Fully Cleared & Verified Credential

Thank You!

Questions & Discussions



Pratyush Khatri



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IILM University, Greater Noida